

Zadanie 31

Oblicz granicę $\lim_{n \rightarrow \infty} \frac{n \sin(n+3)}{(3n-1)^2}$.

① Korzystamy z twierdzenia o trzech ciągach.

$$-1 \leq \sin(n+3) \leq 1$$

$$c_n = \frac{n}{(3n-1)^2} > 0 \quad a_n = \frac{-n}{(3n-1)^2} < 0 \quad b_n = \frac{n \sin(n+3)}{(3n-1)^2}$$

$$\frac{-n}{(3n-1)^2} \leq \frac{n \sin(n+3)}{(3n-1)^2} \leq \frac{n}{(3n-1)^2}$$

$$\text{Zatem } \lim_{n \rightarrow \infty} b_n = 0$$

$$\text{Odp.: } \lim_{n \rightarrow \infty} \frac{n \sin(n+3)}{(3n-1)^2} = 0.$$